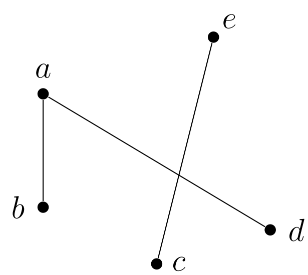
Zack Abdirashid
2/3/26
MATH 405

<u>1.3</u>

6. If a graph G has n vertices, all of which but one have odd degree, how many vertices of odd degree are there in $\overline{G}$, the complement of G?

Since 1 vertex is of even degree, there are $n-1$ vertices of odd degree, implying $n-1$ is an even number. $\therefore n$ must be odd. Let $a \in G$ such that $\deg(a) = 2$.

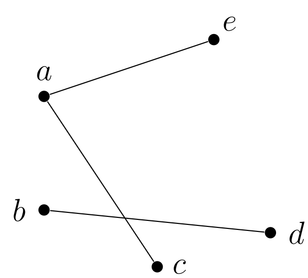
$\underline{G}$:



* "complement" $\Rightarrow$ same # of vertices but $\nexists$ edges from G in $\overline{G}$.

$\underline{\overline{G}}$:

$\Rightarrow$



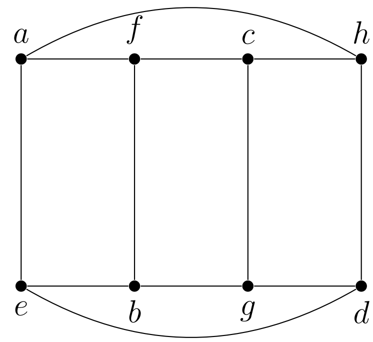
$\Rightarrow$ $\boxed{n-1 \text{ vertices of odd degree}}$

10. There used to be 26 football teams in the National Football League (NFL) with 13 teams in each of two conferences. An NFL guideline said that each team's 14-game schedule should include exactly 11 games against teams in its own conference and three games against teams in the other conference. By considering the right part of a graph model of this scheduling problem, show that this guideline can not be satisfied!

Let C be a graph of one of the conferences containing 13 teams. Representing teams as vertices and edges as games between two teams, there are 13 (odd) vertices each of degree 11 (odd) in C. But, the # of vertices of odd degree in any graph G must be even. $\therefore$ it is impossible for 13 teams to play 11 games each against each other which does not satisfy the guideline.

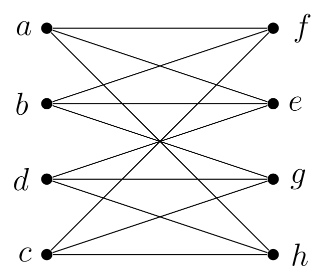
16. Determine whether the following graphs are bipartite. If so, give the partition into left and right vertices.

(a)



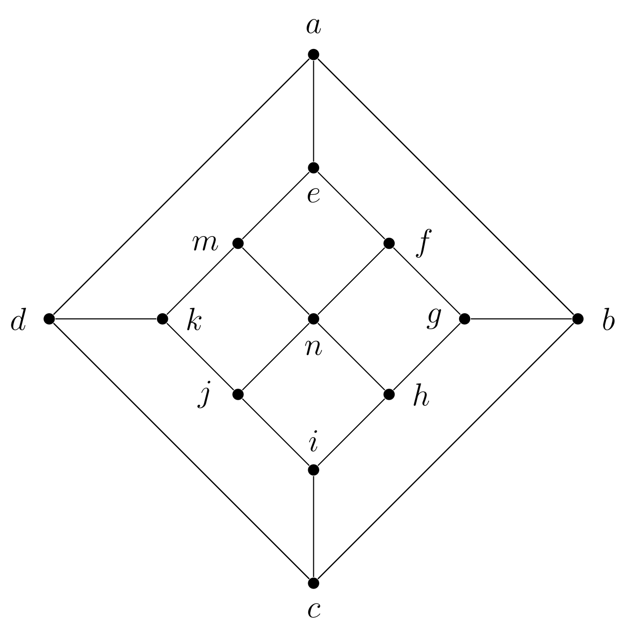
Select vertex a and put it on the left.
* If a has a path of odd length to some vertex x, put x on the right. Otherwise, put x on the left.
* If two adjacent vertices x and y are on the same side, the graph is <u>not</u> bipartite.

$\Rightarrow$

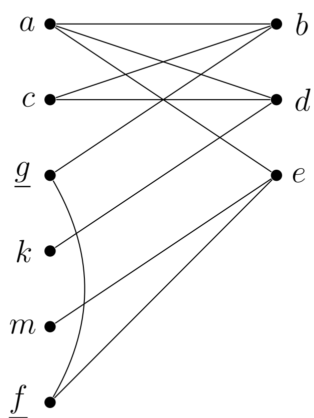


$\Rightarrow$ $\boxed{\text{This graph is bipartite}}$.

(b)



$\Rightarrow$



$\Rightarrow$ $\boxed{\text{This graph is \underline{not} bipartite}}$.